
Compact USB-powered FM receiver with integrated stereo audio amplification, on-board 3.3 V regulation and user control interface.

1 Features

- Integrated FM receiver based on the RDA5807FP
- Full commercial FM broadcast band coverage (87.5 MHz to 108 MHz)
- USB-powered (5 V) operation via USB Type-C connector (VBUS)
- On-board 3.3 V voltage regulator (AMS1117-3.3) dedicated to RF and digital sections
- Integrated stereo audio output stage based on the TDA2822 low-voltage audio amplifier
- Audio amplification optimized for single-supply 5 V operation
- Analog volume control using precision trim potentiometers (3386F series)
- Stereo audio output available via 3.5 mm audio jack or dedicated speaker terminals
- On-board power status LED indicator
- Compact PCB layout suitable for educational use, prototyping and DIY projects
- Selectable audio output via on-board mechanical switch, allowing routing between headphone output and amplified speaker output

2 Applications

- USB-powered FM radio receivers for educational and laboratory use
- DIY radio projects and hobbyist electronics platforms
- Demonstration boards for RF reception, audio processing and amplification
- Portable or desktop FM radio prototypes powered from USB sources

3 Description

The USB FM Radio Receiver & Audio Amplifier Module is a compact, USB-powered platform that integrates FM broadcast reception and audio amplification in a single PCB. The design combines the RDA5807FP FM receiver IC with a low-voltage stereo audio amplifier stage based

on the TDA2822, providing a complete solution for FM radio reception, audio conditioning and output.

The module is powered directly from the USB VBUS (5 V) through a USB Type-C connector. An on-board AMS1117-3.3 linear regulator generates a stable 3.3 V supply dedicated to the RF and digital sections of the circuit, ensuring proper operation of the RDA5807FP while isolating sensitive RF blocks from supply variations. Bulk and ceramic decoupling capacitors are distributed across the board to improve power integrity and reduce noise.

The RDA5807FP performs all FM radio front-end functions internally, including RF amplification, frequency synthesis, demodulation and stereo decoding. The device is controlled via a digital interface (SDA and SCLK), enabling functions such as frequency tuning, automatic seek, mute and preset management. User interaction is provided through on-board push buttons (KEY1...KEY5).

An on-board output selection switch allows the user to route the audio signal either directly to the headphone output or to the TDA2822 low-voltage audio amplifier stage, which is optimized for single-supply 5 V operation. This configuration allows the module to directly drive headphones or small loudspeakers without the need for external amplification stages.

The module is intended primarily for educational purposes, prototyping and DIY projects, serving as a practical reference design for FM radio reception. It is not designed or certified for safety-critical, automotive or medical applications.



Figure 1. Prototype

PCB size
76.36mm X 50mm

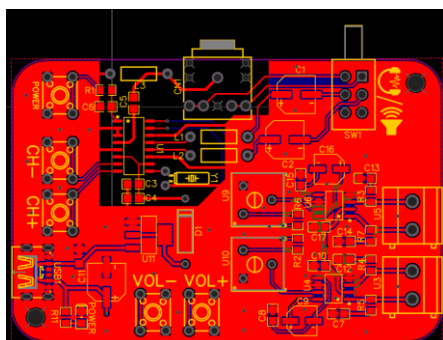


Figure 2. Top Layer

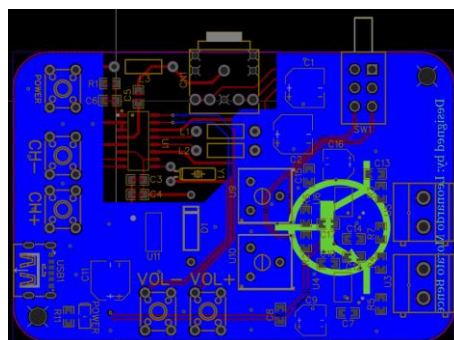


Figure 3. Bottom Layer

Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
USB supply voltage (VBUS)		4.5	5.0	5.5	V
Regulated voltage (VREG)	AMS1117-3.3, IOUT ≤ 200 mA	3.10	3.30	3.45	V
Total supply current (idle)	FM tuned, audio muted	-	35	60	mA
Total supply current (headphone mode)	Volume mid, no speaker load	-	45	80	mA
Total supply current (speaker mode)	RL = 4 Ω, medium volume	-	300	600	mA
FM frequency range		87.5	-	108	MHz
FM receiver sensitivity	SNR = 30 dB	-	2.0	-	μV
Audio output level (headphone)	RL = 32 Ω, THD < 1%	-	120	-	mVrms
Output power per channel (speaker)	BTL, RL = 4 Ω, THD = 10%	-	0.35	0.5	W
Total harmonic distortion (THD)	1 kHz, RL = 4 Ω	-	0.5	10	%
Signal-to-noise ratio (audio)	FM reception, A-weighted	-	50	-	dB
Operating temperature range		-20	-	85	°C
Storage temperature range		-40	-	125	°C

Notes:

- Typical values are based on RDA5807FP and TDA2822 datasheets.
- Actual performance depends on antenna quality.
- USB current consumption must comply with the capabilities of the power source.
- Maximum THD specified at rated output power.

BOM (Bill of Materials)

Name	Designator	Quantity
10nF 50V	C8,C15	2
10uF 50V	C9,C16	2
100uF 16V	C11	1
PJ-307N5	CN1	1
UF4007	D2	1
K4-6x6_TH	KEY1,KEY2,KEY3,KEY4,KEY5	5
1u	L1,L2	2
0.1u	L3	1
LED-0805_B	LED2	1
30k 0805	R1	1
2k2 0805	R2,R6	2
4R7 0805	R3,R4,R5,R7	4
1k 0805	R11	1
PS-22F91	SW2	1
RDA5807FP	U1	1
KF128-5.0-2P	U3,U5	2
TDA2822D	U4,U6	2
3386F-1-103TLF	U9,U10	2
AMS1117-3.3V	U11	1
TYPE-C-6P-DIP2X2	USB1	1
CRYSTAL	Y1	1

Block Diagram

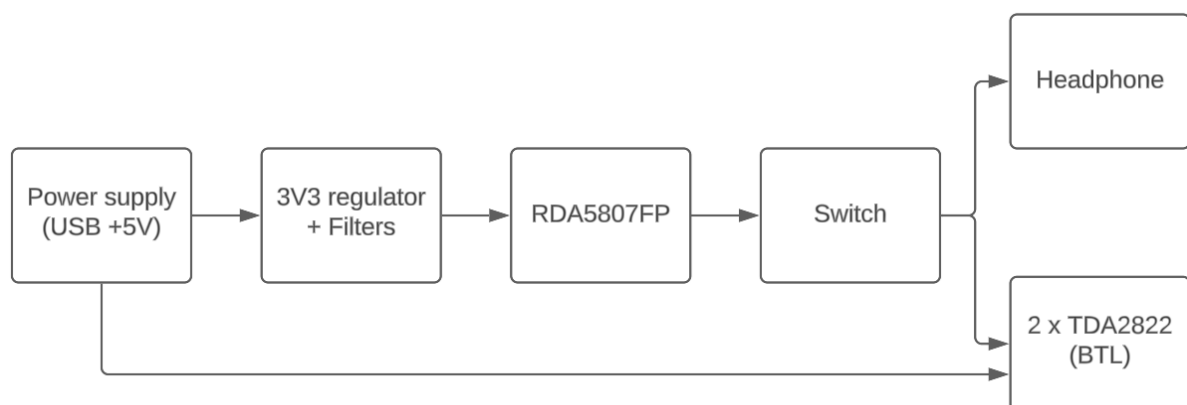


Figure 4. Block Diagram

Schematic

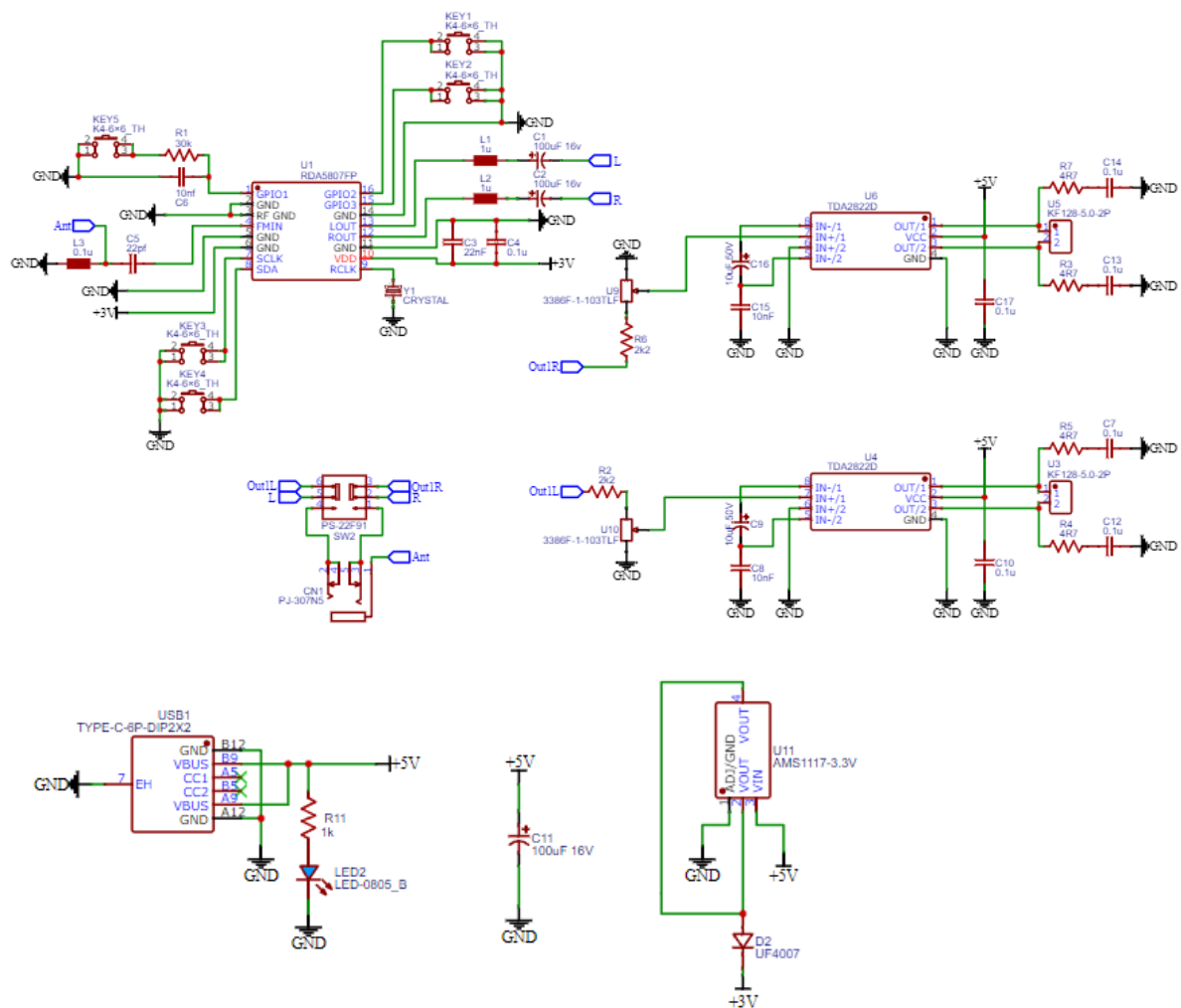


Figure 5. Schematic

3D image of the PCB

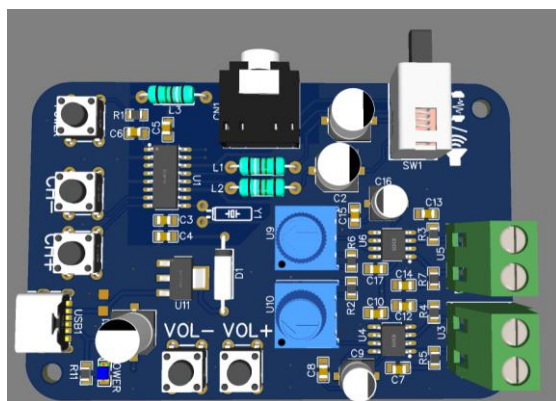


Figure 6. PCB Top Side

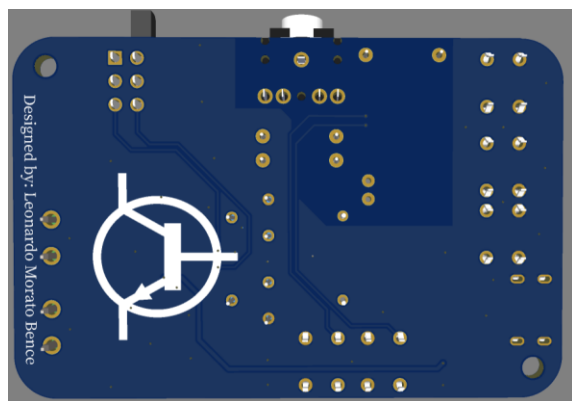


Figure 7. PCB Bottom Side

Disclaimer

The information contained in this document is provided “as is” and is subject to change without notice. This product is intended solely for educational, prototyping, and demonstration purposes. It is not certified for use in safety-critical, medical, automotive, or industrial applications. The author assumes no responsibility for any damages, direct or indirect, resulting from the improper or incorrect use of this module.

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